Roomify

Furniture E-Commerce Platform

PHASE 1 — PROJECT PLANNING & MANAGEMENT

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1. Project Proposal

1.1 Project Overview

Roomify is a direct-to-consumer furniture e-commerce platform built for a single manufacturer. It eliminates marketplace intermediaries, giving the brand full control over the customer experience from product discovery through delivery. The platform's defining feature is an in-browser 3D model viewer — allowing customers to inspect furniture from every angle and, on supported devices, preview it through an augmented reality (AR) camera overlay placed directly in their living space.

1.2 Primary Objectives

- Build a fully functional furniture e-commerce storefront enabling users to browse, filter, search, and purchase products end-to-end.
- Implement an in-browser 3D viewer for furniture models, loading within 3 seconds on supported devices.
- Reduce purchase hesitation through spatial product previews, targeting a 20–30% reduction in return rates vs. baseline.
- Deliver a responsive, mobile-first UI fully functional on iOS Safari, Android Chrome, and desktop browsers.

1.3 Secondary Objectives

- Provide a user account system with order history and saved items (Wishlist).
- Allow users to save and share AR snapshots of furniture in their space.
- Support an admin panel for managing product catalogue and 3D model assets.

1.4 Project Scope

#	In-Scope Feature
1	3D model viewer + AR camera overlay (WebGL / Three.js)
2	Payment gateway integration (Stripe)
3	User authentication (JWT / OAuth)
4	Shopping cart CRUD
5	Product search, filters, and categories
6	Basic backend REST API
7	Online payment integration
8	Admin dashboard (product, order, user management)
9	Delivery tracking web page

1.5 Out of Scope

- Mobile application (iOS / Android native)
- AI recommendations system (deferred to v2)

1.6 Limitations

Limitation	Details
Real-Time Inventory	Live warehouse inventory sync is out of scope. Stock levels will be updated manually or via scheduled batch updates.
Multi-Currency / Language	Initial release supports English only and a single base currency. Localization is a future phase.
AI Recommendations	Personalized AI-driven recommendations are out of scope for v1; basic related-products logic will be used instead.

2. Project Plan

2.1 Timeline Overview

The project runs from 19 April 2025 to 10 July 2025 — 12 weeks in total. Work is organised into five sequential phases with clear handoff criteria between each.

Phase	Name	Start	End	Duration	
Phase 1	Design & Database	W1 (19 Apr)	W1 (25 Apr)	**1 week**	
Phase 2	Backend Development	W2 (26 Apr)	W3 (09 May)	**2 weeks**	
Phase 3	Frontend Development	W4 (10 May)	W8 (06 Jun)	**5 weeks**	
Phase 4	Integration & QA	W9 (07 Jun)	W11 (20 Jun)	**2 weeks**	
Phase 5	Deployment & Launch	W11 (21 Jun)	W12 (10 Jul)	**~2 weeks**	

2.2 Gantt Chart — Task Schedule

Each task is color-coded by phase. All tasks in Phase 3 (Frontend) run in parallel tracks — two frontend developers working concurrently.

Task	Sub-task	Week	Phase	Owner
UI & UX design	Wireframes & design system	W1	**Phase 1**	Moatasem / Mahmoud

Database schema	All tables & relations	W1	**Phase 1**	Abdalrahman / Omar
User auth	JWT / OAuth endpoints	W2	**Phase 2**	Moatasem, Abdalrahman
Product & category API	CRUD + search	W2	**Phase 2**	Moatasem, Abdalrahman
Shopping cart API	Add/remove/update	W2	**Phase 2**	Moatasem, Abdalrahman
Order & payment logic	Stripe integration	W3	**Phase 2**	Moatasem, Abdalrahman
Admin dashboard API	Inventory & orders	W3	**Phase 2**	Moatasem, Abdalrahman
Product listing & search	Filter, sort, categories	W4	**Phase 3**	Abdalrahman
Product detail page	Images, specs	W4	**Phase 3**	Moatasem
3D model viewer	WebGL / Three.js	W5	**Phase 3**	Omar
AR camera feature	Device camera overlay	W5	**Phase 3**	Mahmoud
Shopping cart UI	Cart CRUD, persistence	W6	**Phase 3**	Abdalrahman
Checkout & payment UI	Stripe elements	W6	**Phase 3**	Moatasem
User auth pages	Login, register, profile	W7	**Phase 3**	Omar
Admin dashboard UI	Products, orders, users	W7	**Phase 3**	Mahmoud
Delivery tracking page	Order status & map	W8	**Phase 3**	Abdalrahman
FE / BE integration	Connect all endpoints	W9	**Phase 4**	All Team
Payment gateway E2E test	Full Stripe flow	W9	**Phase 4**	All Team
3D/AR compatibility test	Cross-browser & device	W9	**Phase 4**	All Team
Performance & security	Load test, pen test	W10	**Phase 4**	All Team
UAT & bug fixing	Stakeholder sign-off	W10–W11	**Phase 4**	All Team
CI/CD & infra setup	Docker, cloud deploy	W11	**Phase 5**	All Team
Production deployment	Go-live checklist	W12	**Phase 5**	All Team
Post-launch monitoring	Analytics & errors	W12	**Phase 5**	All Team

2.3 Milestones & Deliverables

#	Date	Milestone	Key Deliverables
M1	W1 (25 Apr)	**Design & DB complete**	Signed mockups, DB schema deployed, design system ready
M2	W3 (09 May)	**Backend APIs complete**	All REST APIs built, Stripe integrated, API docs published
M3	W5 (23 May)	**3D/AR viewer complete**	Three.js viewer live, AR overlay working, browser-compat tested
M4	W8 (06 Jun)	**Frontend complete**	All UI pages built and connected to mock/real APIs
M5	W11 (20 Jun)	**UAT signed off**	E2E tests passing, security pen test done, stakeholder sign-off
M6	W12 (10 Jul)	**Go live**	Production deployed, monitoring active, post-launch runbook delivered

3. Task Assignment & Roles

3.1 Team Members & Responsibilities

Moatasem/ Mahmoud — UI/UX Designer *(W1–W8)*

Lead wireframes, high-fidelity mockups, design system, 3D/AR interaction prototyping,

Omar / Abdalrahman — Database Architect *(W1)*

Design and implement database schema: products, users, orders, cart, 3D assets, reviews. Write migration scripts. Document the schema.

Moatasem / Abdalrahman — Backend Developer *(W2–W5)*

Build all REST APIs (auth, products, cart, orders, admin). Stripe payment integration. Admin dashboard API. API documentation (Swagger).

All Frontend Devs — Frontend Developers *(W4–W12)*

Build all UI pages (listing, detail, 3D viewer, AR, cart, checkout, auth, admin, delivery). FE/BE integration. Cross-browser testing.

4. Risk Assessment & Mitigation Plan

4.1 Risk Register

R1 3D/AR implementation failure Probability: **High** Impact: **Critical** Owner: Moatasem M.

Description: Team has limited WebGL/Three.js experience. AR camera feature may not work across all browsers/devices.

Mitigation: Fallback: deliver 3D viewer only (no AR) if AR proves infeasible. Start Three.js prototyping in W1 as a parallel spike. Use model-viewer web component as a backup for AR.

Trigger: *No working 3D prototype by end of W2*

R2 Stripe integration delays Probability: **Medium** Impact: **High** Owner: Moatasem A.

Description: Payment gateway setup requires business verification, API key approval, and compliance steps that can take days to weeks.

Mitigation: Apply for Stripe account in W1. Use Stripe test mode throughout development. Have a documented fallback to manual payment (COD) for launch if approval is delayed.

Trigger: *No Stripe approval by W8*

R3 3D model assets not ready Probability: **Medium** Impact: **High** Owner: Moatasem M.

Description: 3D furniture model files (.glb/.gltf) may not be available or optimised in time for frontend development.

Mitigation: Source or create placeholder 3D models in W1. Set a hard deadline with asset owners for final models by W4. Define a maximum file-size budget (< 5 MB per model).

Trigger: *No approved 3D assets by W3*

R4 Scope creep Probability: **High** Impact: **Medium** Owner: Moatasem M.

Description: Stakeholders may request additional features mid-project, threatening the 12-week deadline.

Mitigation: Maintain a locked scope document. Any new feature requests go into a v2 backlog. Weekly status meetings with explicit scope-change approval process.

Trigger: *Feature request not in signed scope*

R5 Team bandwidth and availability Probability: **Medium** Impact: **High** Owner: All

Description: Small team (4 members) with limited redundancy. Any absence on a critical path task can delay the timeline.

Mitigation: Cross-train on critical paths (FE devs understand API contracts). Document all work. Two-day buffer built into each phase handoff. Maintain a shared task board.

Trigger: *Any team member unavailable > 2 days on critical path*

R6 Performance targets not met Probability: **Low** Impact: **High** Owner: Moatasem A.

Description: Page load > 2s or API response > 300ms may result in poor UX and failed KPIs.

Mitigation: Establish performance budgets in W1. Run Lighthouse in CI pipeline. Lazy-load 3D models. Optimise API queries in W9 integration phase.

Trigger: *Lighthouse score < 80 in W10 audit*

5. KPIs — Key Performance Indicators

5.1 KPI Summary

12 KPIs are defined across 4 categories: Performance, Reliability, Business, and User Experience.

Critical KPIs must be met before go-live (10 July 2025). High KPIs must be met within 30 days post-launch.

All KPIs will be reviewed weekly during Phase 4 (Integration & QA) and monthly post-launch.

PERFORMANCE KPIs

KPI-01 Page Load Time Target: **< 2 seconds** Priority: **Critical**

Time for any page to become interactive from initial request (Largest Contentful Paint).
How: Google Lighthouse / Core Web Vitals
Tool: Lighthouse, PageSpeed Insights
Frequency: Every sprint + every page
Rationale: *Each 1s delay reduces conversions by ~7% (Google research).*

KPI-02 API Response Time (p95) Target: **< 300 ms** Priority: **Critical**

Server response time for all REST API calls at the 95th percentile.
How: Backend APM — measure time from request receipt to response sent
Tool: Datadog / New Relic / Prometheus
Frequency: Continuous monitoring
Rationale: *Slow APIs block frontend rendering and cart operations directly.*

KPI-03 3D Model Load Time Target: **< 5 seconds** Priority: **High**

Time to fully render a 3D furniture model in the browser viewer (first model load).
How: Custom JS performance.Now() timer from load start to render complete event
Tool: Custom instrumentation + Sentry
Frequency: Per model, per browser, post-launch
Rationale: *High abandonment rates observed on product detail pages when 3D load exceeds 5s.*

KPI-04 Search Response Time Target: **< 500 ms** Priority: **High**

Time from search query submission to results rendered on screen.
How: Frontend timing events + backend query execution time logs
Tool: Backend logs + frontend analytics
Frequency: Weekly QA + post-launch
Rationale: *Slow search = users give up before seeing results.*

RELIABILITY KPIs

KPI-05 System Uptime Target: **99.9% monthly** Priority: **Critical**

Percentage of time the website and all APIs are fully available to end users.
How: External uptime monitoring with 1-minute polling intervals
Tool: UptimeRobot / Better Uptime / AWS CloudWatch
Frequency: Continuous, reported monthly
Rationale: *99.9% = max 44 min downtime per month. Any more = measurable revenue loss.*

KPI-06 Error Rate Target: **< 0.5% of requests** Priority: **Critical**

Percentage of HTTP requests that result in a 5xx server error.

How: Error count ÷ total requests × 100, tracked in error monitoring tool
Tool: Sentry / Datadog
Frequency: Continuous, alert on spike > 1%
Rationale: *High error rate = broken checkouts, failed API calls, lost orders.*

KPI-07 Payment Success Rate Target: **> 98%** Priority: **Critical**
Percentage of checkout attempts that complete successfully via Stripe.
How: Successful orders ÷ initiated checkouts × 100
Tool: Stripe Dashboard + internal order database
Frequency: Daily review, alert if < 97%
Rationale: *Direct lost revenue metric — every failed payment is a lost sale.*

BUSINESS KPIs

KPI-08 Conversion Rate Target: **> 2.5%** Priority: **High**
Percentage of unique website visitors who complete a purchase.
How: Orders ÷ unique sessions × 100
Tool: Google Analytics 4 + internal order DB
Frequency: Weekly from launch
Rationale: *Industry benchmark for furniture e-commerce: 2–3%. Primary revenue health indicator.*

KPI-09 Cart Abandonment Rate Target: **< 65%** Priority: **High**
Percentage of users who add items to cart but do not complete checkout.
How: 1 − (completed checkouts ÷ carts created) × 100
Tool: GA4 funnel analysis + internal cart DB
Frequency: Weekly from launch
Rationale: *Industry average ~70%. Reduce via saved carts, guest checkout, and streamlined UX.*

USER EXPERIENCE KPIs

KPI-10 User Adoption Rate Target: **> 40% returning** Priority: **High**
Percentage of users who return to the site after their first visit (within 30 days).
How: Returning users ÷ total users × 100
Tool: Google Analytics 4 audience report
Frequency: Monthly
Rationale: *Indicates product–market fit and brand loyalty for a D2C furniture brand.*

KPI-11 3D/AR Feature Engagement Target: **> 35% of product page visitors** Priority: **High**
Percentage of product detail page visitors who interact with the 3D viewer or AR feature.
How: Custom event tracking on viewer open + AR session start
Tool: GA4 custom events / Mixpanel
Frequency: Weekly post-launch
Rationale: *Core product differentiator — low adoption signals UX or discoverability issues.*

DEVELOPMENT KPIs

KPI-12 Bug Escape Rate Target: **< 5% to production** Priority: **Medium**
Percentage of total bugs found in production versus bugs caught during QA.
How: Production bugs ÷ (QA bugs + production bugs) × 100
Tool: Sentry + Jira / Linear bug tracking
Frequency: Per release / sprint
Rationale: *High escape rate indicates weak test coverage or rushed QA cycle.*